Malware Analysis

Slip 1:

Q1. Steps to Download and Install VirtualBox (Windows)

✅ **Prerequisites (Before Installation)**

1. Your system should be **64-bit Windows (Windows 10/11)**
2. Minimum **4 GB RAM** (8 GB recommended)
3. At least **20 GB free disk space**
4. Enable **Virtualization (VT-x/AMD-V)** in BIOS
5. Disable Hyper-V (if causing issues)

▼ **Download Link**

- Official website:
  👉 https://www.virtualbox.org
  (Click on **"Downloads"** → **Windows hosts**)

🛠 **Installation Steps**

1. Go to the website and download VirtualBox for Windows
2. Open the downloaded .exe file
3. Click **Next** on setup wizard
4. Select installation location (default is fine)
5. Click **Next** → **Yes** → **Install**
6. Allow permissions if asked
7. Wait for installation to complete
8. Click **Finish**

⚙️ After Installation

1. Open VirtualBox
2. Click **"New"** to create a virtual machine
3. Select OS type (Windows/Linux)
4. Allocate RAM & storage
5. Start using VM

Q2. Using Process Hacker / Process Monitor for Malware Analysis

🎯 Goal

Track:

- Multiple processes
- Suspicious activity
- Network connections

🔍 Using Process Hacker

Steps:

1. Open **Process Hacker (Run as Administrator)**
2. Look at **running processes list**
3. Check for:
    - Unknown or suspicious process names
    - High CPU or memory usage
4. Right-click suspicious process →
    - **Properties** → check path, behavior
    - **Threads/Handles** → see activity
5. Use **"Network" tab**:
    - Check unusual IP connections

6. Kill or suspend process (for testing)

🔍 Using Process Monitor (ProcMon)

Steps:

1. Open **Process Monitor**
2. Click **Start Capture (Ctrl + E)**
3. Filter data:
   - Filter by **Process Name**
4. Observe activities:
   - File access (Read/Write/Delete)
   - Registry changes
   - DLL loading
5. Look for:
   - Suspicious file paths
   - Unknown registry entries
6. Stop capture when done

⚠️ What to Look For (Important)

- Processes creating **multiple child processes**
- Unknown **.exe files in temp folders**
- Frequent **network connections to unknown IPs**
- Changes in **registry (auto-start entries)**

✅ Conclusion (Simple)

- **Process Hacker** → good for real-time process + network monitoring

- **Process Monitor** → detailed logs of file & registry activity
- Together, they help **detect and analyze malware behavior**

Slip 2:

Q1. Steps to Download and Install VMware (Windows)

✅ **Prerequisites (Before Installation)**

1. **64-bit Windows (Windows 10/11)**
2. Minimum **4 GB RAM** (8 GB recommended)
3. At least **20 GB free disk space**
4. **Virtualization enabled** in BIOS (Intel VT-x / AMD-V)
5. Disable **Hyper-V** (to avoid conflicts)

▼ **Download Link**

- Official website:
  👉 https://www.vmware.com
- Go to **Products** → **Workstation Player (Free)**

🛠 **Installation Steps**

1. Visit the website and download **VMware Workstation Player**
2. Open the downloaded .exe file
3. Click **Next** on setup wizard
4. Accept the license agreement
5. Choose installation location (default is fine)
6. Click **Next** → **Install**
7. Wait for installation to finish

8. Click **Finish** and restart system if asked

⚙️ After Installation

1. Open VMware
2. Click **Create New Virtual Machine**
3. Select OS (Windows/Linux ISO file)
4. Allocate RAM, CPU, and storage
5. Start the virtual machine

Q2. Documentation & Reporting of Malware Analysis

After using tools like **Noriben**, **Process Monitor**, and **Process Hacker**, you should create a **clear and structured report**.

📄 Key Elements of Analysis Report

1. Title Page

- Report title
- Your name
- Date
- Sample name (malware file)

---

2. Objective

- Purpose of analysis
- Example: “To analyze behavior of suspected malware sample”

---

3. Tools Used

- Noriben

- Process Monitor
- Process Hacker

(Write version if possible)

---

4. Sample Information

- File name
- File type (.exe, .dll)
- File size
- Hash value (MD5/SHA256)

---

5. Environment Details

- OS used (Windows version)
- VM details (RAM, CPU)
- Network setup (isolated/sandbox)

---

6. Dynamic Analysis Findings

◈ Process Activity

- New processes created
- Parent-child process relation
- Suspicious process names

◈ File Activity

- Files created/deleted
- Location (Temp, AppData, etc.)

◈ Registry Changes

- New registry keys
- Auto-start entries

◈ Network Activity

- IP addresses contacted
- Suspicious domains
- Ports used

---

7. Screenshots / Logs

- Add screenshots from:
  - Process Hacker
  - Process Monitor
  - Noriben logs

---

8. Indicators of Compromise (IOCs)

- File hashes
- Suspicious file paths
- IP addresses / domains

---

9. Conclusion

- Final result:
  - Is it malware or safe?
  - Type of malware (if known)

---

10. Recommendations

- Block IPs/domains
- Delete infected files
- Use antivirus/firewall
- Update system security

✅ Simple Conclusion

- Write report in **clear sections**
- Include **process, file, registry, and network details**
- Add **screenshots + logs** for proof
- Highlight **important threats (IOCs)**

Slip 3:

Q1. Steps to Install IDA Pro on Windows

✅ Prerequisites (Before Installation)

1. **64-bit Windows (Windows 10/11)**
2. Minimum **4 GB RAM** (8 GB recommended)
3. At least **5–10 GB free disk space**
4. Admin rights on system
5. Download from official source

▼ Download Link

- Official website:
  👉 https://hex-rays.com

🛠 Installation Steps

1. Go to the website and download **IDA Pro setup file**
2. Extract the downloaded .zip file (if compressed)
3. Run the installer (setup.exe) as Administrator
4. Click **Next** on setup wizard
5. Accept the license agreement
6. Choose installation location

7. Select components (keep default)
8. Click **Install**
9. Wait for installation to complete
10. Click **Finish**

⚙ Configuration (After Installation)

1. Open IDA Pro
2. Activate license (if required)
3. Set default directories for projects
4. Load a sample file (.exe, .dll)
5. Choose processor type (auto-detected usually)
6. Let IDA analyze the file

⚠ Important Notes

- Use **legal licensed version**
- Disable antivirus temporarily if blocking install
- Keep system updated

Q2. Key Configuration Settings Before Creating a VM

After installing **VMware / VirtualBox**, check these settings:

⚙ 1. Virtualization Support

- Ensure **VT-x / AMD-V is enabled in BIOS**
- Without this, VM will be slow or not run

💾 2. RAM Allocation

- Decide how much RAM to give VM
- Example:
  - Host has 8 GB → give VM 2–4 GB

---

3. CPU Cores

- Allocate CPU cores properly
- Example:
  - 4-core CPU → give VM 2 cores

💽 4. Storage Settings

- Choose disk type:
  - Dynamically allocated (recommended)
- Set enough space (20–40 GB)

🌐 5. Network Settings

- NAT → Internet access (safe)
- Host-only → isolated (for malware analysis)

💿 6. ISO File / OS Setup

- Attach OS ISO file (Windows/Linux)
- Ensure file is valid and bootable

🔒 7. Security Settings

- Disable shared folders (for malware testing)
- Disable clipboard sharing if needed

🗀 8. Snapshot Feature

- Enable snapshots
- Useful to restore system after testing

🖥 9. Display Settings

- Allocate video memory
- Enable 3D acceleration if needed

⚠ 10. USB & Device Settings

- Disable unnecessary USB devices
- Prevent malware spreading

✅ Simple Conclusion

- Always check **RAM, CPU, storage, and network** before creating VM
- Use **host-only network + snapshots** for safe malware analysis
- Proper configuration = **better performance + safety**

Slip 4:

Q1. Steps to Create a New Linux VM in VirtualBox

🖥 Step-by-Step Process

1. Open VirtualBox

- Start VirtualBox application
- Click on **“New”**

---

2. Enter VM Details

- Name: (e.g., Ubuntu Linux)
- Type: **Linux**
- Version: Select your Linux version (Ubuntu, Kali, etc.)

---

3. Allocate RAM (Memory)

- Minimum: **2 GB (2048 MB)**
- Recommended: **4 GB or more**
  ☞ Do not use more than half of your system RAM

---

4. Allocate CPU Cores

- Go to **Settings** → **System** → **Processor**
- Assign:
  - 1–2 cores (basic use)
  - 2–4 cores (better performance)

---

5. Create Virtual Hard Disk

- Select: **Create a virtual hard disk now**
- Disk type: **VDI (VirtualBox Disk Image)**
- Storage: **Dynamically allocated**

---

6. Allocate Disk Space

- Minimum: **20 GB**
- Recommended: **30–50 GB**

---

7. Attach ISO File

- Go to **Settings** → **Storage**

- Add Linux ISO file (Ubuntu/Kali)

---

8. Start the VM

- Click **Start**
- Follow Linux installation steps

⚙️ Tips for Optimal Performance

- Enable **VT-x / AMD-V (Virtualization)**
- Enable **PAE/NX** (in processor settings)
- Increase **Video Memory (128 MB)**
- Enable **3D Acceleration**
- Use **Guest Additions** for better speed

✅ Conclusion

- Proper RAM, CPU, and disk allocation ensures **smooth performance**
- Always balance resources between **host and VM**

Q2. Opening and Analyzing a Malicious DLL in IDA Pro

🛠️ Steps to Open DLL in IDA Pro

1. Open IDA Pro
2. Click **File** → **Open**
3. Select the malicious .dll file
4. Choose:
    - File type: Portable Executable (auto-detected)
5. Click **OK**

6. Wait for auto-analysis to complete

🔍 Initial Analysis Steps

1. Identify Entry Point

- DLL does not use main()
- Look for:
  - **DllMain()** (main function of DLL)

---

2. Check Exports

- Go to **Exports window**
- See exported functions (used by other programs)

---

3. Analyze Functions

- Open **Functions window**
- Look for suspicious names or patterns

---

4. Look for Dangerous APIs

Check for API calls like:

- File operations (CreateFile, WriteFile)
- Registry changes (RegSetValue)
- Network activity (connect, send)
- Process injection (CreateRemoteThread)

---

5. Strings Analysis

- Open **Strings window**
- Look for:

    - URLs
    - IP addresses
    - File paths
    - Suspicious commands

---

6. Cross References (XREF)

- See where functions are used
- Helps track malware flow

---

7. Check Imports

- Go to **Imports section**
- Identify Windows APIs used by DLL

⚠ **Indicators of Malicious Behavior**

- Hidden or obfuscated code
- Calls to network APIs
- Auto-start registry entries
- Code injection techniques

✅ **Conclusion**

- Start with **DllMain + exports + strings**
- Focus on **API calls and suspicious behavior**
- Use IDA Pro to understand **malware logic step-by-step**

Slip 5:

Q1. Steps to Create Windows 10 VM in VMware Workstation Player

📀 Step-by-Step Process

1. Open VMware Workstation Player

- Launch VMware
- Click **“Create a New Virtual Machine”**

---

2. Select Installation Method

- Choose **Installer disc image file (ISO)**
- Browse and select **Windows 10 ISO file**

---

3. Enter Windows Details

- Enter:
  - Product key (optional)
  - Username & password

---

4. Name and Location

- Give VM name (e.g., Windows 10 VM)
- Choose storage location

---

5. Allocate Disk Space

- Minimum: **40 GB**
- Recommended: **60 GB+**
- Select:
  - **Split disk into multiple files** (recommended)

⚙️ Configure System Resources

6. Customize Hardware (Important Step)

Click **“Customize Hardware”**

🖫 RAM (Memory)

- Minimum: **4 GB**
- Recommended: **8 GB**

▯ CPU Cores

- Example:
  - 4-core system → give **2 cores**
- Avoid using all cores

💿 Hard Disk

- Ensure enough storage (40–60 GB)
- Use **SSD (if available)** for better performance

🌐 Network

- NAT → normal internet use
- Host-only → safe malware testing

🖥 Display

- Increase video memory
- Enable **Accelerate 3D Graphics**

⊙ CD/DVD

- Ensure ISO file is attached

---

7. Finish Setup

- Click **Finish**
- Start VM and install Windows 10

⚠️ **Performance Tips**

- Enable **Virtualization in BIOS**
- Install **VMware Tools**
- Close unnecessary host apps

✅ **Conclusion**

- Allocate **balanced CPU, RAM, and disk**
- Proper setup ensures **smooth VM performance**

Q2. IDAPython Script to Extract Functions

□ Simple Script (List All Functions)

```
import idautils

for func in idautils.Functions():
    print(hex(func), idc.get_func_name(func))
```

🔍 **Explanation (Simple)**

- idautils.Functions() → gets all functions
- idc.get_func_name() → gets function name
- hex(func) → shows function address

⚠ Modify Script for Suspicious Functions

Example (Filter by Keywords)

```
import idautils
import idc

suspicious_keywords = ["CreateFile", "WriteFile", "Reg", "Socket",
"Connect", "VirtualAlloc", "LoadLibrary"]

for func in idautils.Functions():
    name = idc.get_func_name(func)

    for keyword in suspicious_keywords:
        if keyword.lower() in name.lower():
            print("Suspicious Function:", hex(func), name)
```

🔍 What This Does

- Checks function names for:
  - File operations
  - Registry changes
  - Network activity
  - Memory allocation
- Prints only **suspicious functions**

⚠ Examples of Suspicious APIs

- CreateFile, WriteFile → file activity
- RegSetValue → registry change
- connect, send → network
- VirtualAlloc → memory injection

✅ Conclusion

- First script → lists all functions
- Modified script → filters **malware-related functions**
- Helps in **quick malware analysis**

Slip 6:

Q1. Installing Tools on Linux (xxd, file, md5sum, sha256sum, sha1sum)

✅ Step 1: Update System

sudo apt update

🛠 Install Required Tools

1. Install xxd

sudo apt install xxd

---

2. Install file

sudo apt install file

---

3. Install md5sum, sha1sum, sha256sum

👉 These tools come with **coreutils**, usually pre-installed

If not installed:

sudo apt install coreutils

🔍 Verify Installation

xxd --version
file --version

```
md5sum --version
sha1sum --version
sha256sum --version
```

💡 Usage Example

```
file sample.exe
md5sum sample.exe
sha1sum sample.exe
sha256sum sample.exe
xxd sample.exe
```

✅ Conclusion

- Most tools are **pre-installed in Linux**
- Use apt install if missing
- These tools help in **static malware analysis**

Q2. Downloading Malware Sample & Safety Precautions

⚠️ Step-by-Step Process (Safe Method)

1. Use Trusted Sources

- Download only from:
  - Malware research websites
  - Security labs

---

2. Use Virtual Machine (Important)

- Always download inside **VM (VirtualBox / VMware)**
- Never use your main system

---

3. Disable Internet (Optional)

- Use **Host-only network** for safety
- Prevent malware communication

---

4. Download the Sample

- Use browser or tools like:

wget <malware_link>

---

5. Verify File Hash

sha256sum sample.exe

- Compare hash with source

🔒 **Precautions While Handling Malware**

🚫 **1. Never Run Directly**

- Do not double-click the file

🔒 **2. Use Isolated Environment**

- Use VM or sandbox
- Take **snapshot before testing**

🛑 **3. Disable Sharing**

- Turn off:
  - Shared folders
  - Clipboard
  - Drag & drop

🔍 **4. Use Analysis Tools Only**

- Open with:
  - Static tools (strings, file, IDA)

🌐 5. Block Network (if needed)

- Prevent data leakage

---

6. Rename File Extension (Optional)

- Example:
  - sample.exe → sample.bin
- Prevent accidental execution

🛡️ 7. Use Antivirus Carefully

- Disable temporarily only if needed

💾 8. Backup & Restore

- Use VM snapshot to restore system

⚠️ Important Risks

- Data theft
- System damage
- Network infection

✅ Conclusion

- Always use **VM + trusted source + isolation**
- Never run malware directly

- Follow safety to avoid **system compromise**

Slip 7:

Q1. Tracking Malicious DNS Requests using Wireshark

🛠 **Steps to Track DNS Requests**

1. Open Wireshark

- Start Wireshark
- Select your **network interface** (Wi-Fi/Ethernet)

---

2. Start Capture

- Click **Start Capture**
- Let traffic run while malware executes (in VM)

---

3. Apply DNS Filter

dns

👉 Shows only DNS traffic

---

4. Analyze Packets

- Click on DNS packets
- Check:
  - **Query name (domain)**
  - **Response IP address**

---

5. Follow DNS Stream

- Right-click → **Follow** → **UDP Stream**
- See full communication

🔍 Abnormal Patterns to Look For

⚠ 1. Suspicious Domain Names

- Random or strange domains
- Example: abcxyz123.ru

⚠ 2. Too Many DNS Requests

- Continuous DNS queries (possible malware beaconing)

⚠ 3. Failed DNS Requests

- Many **NXDOMAIN responses**
- Indicates domain generation algorithm (DGA)

⚠ 4. Unusual Geographic Locations

- Domains pointing to unknown or risky countries

⚠ 5. Long or Encoded Domains

- Base64-like strings in domain names

⚠ 6. Repeated Same Domain

- Same request again and again (command & control)

⚠ 7. Suspicious Record Types

- TXT records (used to hide data)

✅ Conclusion

- Use **Wireshark filters + packet analysis**
- Look for **strange domains, repetition, failures**
- Helps identify **malware communication**

Q2. Analyzing Suspicious Function in IDA Pro

🛠 **Using IDA Pro (Manual Analysis)**

1. Open Function

- Go to **Functions window**
- Double-click suspicious function

---

2. Check Assembly Code

- Look at instructions:
  - API calls
  - Loops
  - Conditions

---

3. Identify API Calls

Focus on:

- Network → connect, send, recv
- File → CreateFile, WriteFile
  👉 These indicate malicious behavior

---

4. Use Cross References (XREF)

- Press **X** on function
- See where it is called from
- Understand program flow

---

5. Check Strings

- Press **Shift + F12**
- Find related:
  - URLs
  - File paths

---

6. Rename Variables

- Rename unclear names to meaningful ones
- Example:
  - sub_401000 → send_data_to_server

🤘 Using IDAPython (Automation)

◈ Script to Print Called APIs in Function

```
import idautils
import idc

for func in idautils.Functions():
    name = idc.get_func_name(func)
    for insn in idautils.FuncItems(func):
        disasm = idc.GetDisasm(insn)

        if "connect" in disasm or "send" in disasm or "CreateFile" in disasm:
            print("Suspicious API in:", name)
```

◈ What This Does

- Scans all functions

- Detects:
  - Network activity
  - File operations
- Helps find **malicious functions quickly**

⚠️ Things to Identify

- Data sent to server
- Files created/modified
- Hidden or encrypted data
- Looping network communication

✅ Conclusion

- Use **IDA interactive tools (XREF, strings, APIs)**
- Use **IDAPython scripts for automation**
- Together they help understand **malware behavior deeply**

Slip 8:

Q1. Using md5sum, sha1sum, sha256sum for Hashing

🛠️ Steps to Generate Hash Values

1. Open Terminal (Linux)

---

2. Navigate to File Location

cd /path/to/file

---

3. Generate MD5 Hash

```
md5sum sample.exe
```

---

4. Generate SHA1 Hash

```
sha1sum sample.exe
```

---

5. Generate SHA256 Hash

```
sha256sum sample.exe
```

📌 Output Example

```
MD5:    a1b2c3...
SHA1:   d4e5f6...
SHA256: g7h8i9...
```

🔍 Importance of Hash Values in Malware Analysis

✅ 1. Unique Identification

- Each file has a **unique hash value**
- Works like a fingerprint

✅ 2. Integrity Check

- Ensures file is **not modified**
- Same file → same hash

✅ 3. Malware Detection

- Compare hash with:
  - Virus databases
  - Threat intelligence

✅ 4. Fast Searching

- Analysts search malware using hash
- Saves time

✅ 5. Evidence in Investigation

- Used in **digital forensics reports**
- Proves file authenticity

⚠️ Note

- MD5 and SHA1 are weaker
- **SHA256 is more secure and preferred**

✅ Conclusion

- Use commands to quickly generate hashes
- Hashes help in **identification, verification, and detection**

Q2. Installing x64dbg on Windows

🛠 Steps to Install x64dbg

1. Download x64dbg

- Official website:
  👉 https://x64dbg.com

---

2. Download ZIP File

- Choose latest release
- Usually comes as .zip file

---

3. Extract Files

- Right-click → **Extract Here**

---

4. Run Application

- Open folder
- Run:
  - x64dbg.exe (for 64-bit)
  - x32dbg.exe (for 32-bit)

⚙ Dependencies

- No heavy installation needed
- Portable tool (runs directly)

Optional:

- Microsoft Visual C++ Redistributable (if error occurs)

⚙ Configuration Settings

1. Run as Administrator

- Needed for debugging system processes

---

2. Set Symbol Path (Optional)

- Helps in better analysis

---

3. Configure Plugins

- Add plugins for advanced analysis

---

4. Disable Antivirus (if blocking)

- Some AV tools may block debugger

---

5. Use in Virtual Machine

- Recommended for malware analysis

⚠️ Safety Tips

- Never debug malware on main system
- Use VM with snapshot
- Disable internet if needed

✅ Conclusion

- x64dbg is **easy, portable debugger**
- No complex setup required
- Use proper configuration for **safe malware analysis**

Slip 9:

Q1. Verifying Malware Hash with Antivirus Databases

🛠 Steps to Verify Hash

1. Generate Hash

- Use:
    - md5sum
    - sha1sum
    - sha256sum (preferred)

---

2. Copy the Hash Value

- Example:
  - a1b2c3d4...

---

3. Search on Malware Databases

🔍 **Popular Platforms**

- VirusTotal
- Hybrid Analysis
- MalwareBazaar

---

4. Paste Hash in Search Bar

- Do NOT upload file (for safety)
- Search using hash only

---

5. Analyze Results

- Check:
  - Detection rate (e.g., 50/70 AV engines)
  - Malware name/type
  - Behavior details

🔍 **What If No Results Found?**

- File may be:
  - New (zero-day malware)
  - Not yet detected

👉 Then perform:

- Static analysis

- Dynamic analysis

📌 Importance

- Saves time (no need to analyze known malware)
- Identifies malware family
- Provides threat intelligence

⚠️ Safety Tip

- Avoid uploading sensitive files
- Use isolated system (VM)

✅ Conclusion

- Use hash → search in databases → check detection
- Helps quickly identify **known malware samples**

Q2. Initial Setup in x64dbg & Avoiding Anti-Debugging

🛠 Initial Steps After Opening Binary

1. Open File

- Run x64dbg
- Click **File** → **Open**
- Select malware .exe

---

2. Start Debugging

- Program pauses at **Entry Point**

---

3. Set Breakpoints

- Press **F2** to set breakpoint
- Focus on:
  - Entry point
  - Important APIs

---

4. Step Execution

- Use:
  - **F7 (Step Into)**
  - **F8 (Step Over)**

---

5. Analyze Registers & Memory

- Check:
  - CPU registers
  - Stack
  - Memory values

⚠ Common Anti-Debugging Techniques (Used by Malware)

- Checking debugger presence
- Timing checks
- API tricks like:
  - IsDebuggerPresent()
  - CheckRemoteDebuggerPresent()

🛡 How to Avoid Anti-Debugging

🔧 1. Use Plugins

- Install plugins like:

    - ScyllaHide (very useful)

◈ 2. Patch Instructions

- Modify code:
    - Change jump conditions
    - Skip anti-debug checks

◈ 3. Hide Debugger

- Use built-in options or plugins
- Prevent detection

◈ 4. Set Breakpoints Smartly

- Avoid obvious locations
- Break after anti-debug code

◈ 5. Use VM Environment

- Run debugger inside VM
- Take snapshot before testing

◈ 6. Modify API Return Values

- Force:
    - IsDebuggerPresent() → return FALSE

🔍 Extra Tips

- Watch for:

    - Sudden program exit
    - Infinite loops
    - Crashes (anti-debug sign)

✅ Conclusion

- Setup debugger with **breakpoints + step execution**
- Use plugins and patching to bypass **anti-debugging**
- Helps analyze malware safely and effectively

Slip 10:

Q1. Loading a Malicious DLL in x64dbg

⚠️ Important Note

- DLL files cannot run directly like .exe
- Need a **loader program** (host process)

🛠 Steps to Load DLL in x64dbg

1. Open x64dbg

- Run **x64dbg.exe (64-bit)**

---

2. Load a Host Executable

- Click **File** → **Open**
- Open a program like:
    - rundll32.exe (Windows tool)

---

3. Pass DLL as Argument

- In command line parameters, add:

your.dll,EntryPoint

👉 Example:

malware.dll,DllMain

---

4. Start Debugging

- Click **Run (F9)**
- Debugger will load DLL inside process

---

5. Set Breakpoints

- Set breakpoint on:
  - DllMain
  - Exported functions

---

6. Step Through Code

- Use:
  - F7 (Step Into)
  - F8 (Step Over)

🔍 **Ensure Proper Attachment**

✅ **Check These:**

- DLL is loaded in **Modules tab**
- Breakpoint hits correctly
- No immediate crash

⚠️ **Safety Tips**

- Use **Virtual Machine**
- Disable internet
- Take snapshot before testing

✅ Conclusion

- Use **rundll32.exe or loader** to run DLL
- Set breakpoints and monitor execution
- Ensure debugger is properly attached

Q2. Scanning File using VirusTotal

🛠 Steps to Scan File

1. Open Website

- Go to:
  👉 VirusTotal

---

2. Upload File

- Click **“Choose file”**
- Select suspicious .exe or .dll

---

3. Start Scan

- Click **Upload / Scan**
- Wait for analysis

---

4. View Results

- After scan, results will show

🔍 How to Interpret Results

📊 1. Detection Ratio

- Example:
  - **45/70** → 45 antivirus detected malware
    👉 Higher number = more dangerous

🏷 2. Malware Name

- Shows malware family/type

🌐 3. Behavior Details

- Network connections
- File changes

📄 4. File Information

- Hash values
- File type
- Size

💬 5. Community Comments

- Security researchers' opinions

⚠ Precautions

- Do NOT upload sensitive/private files
- Use **hash search instead of file upload** (safer)
- Perform in **VM environment**

✅ Conclusion

- Upload file → analyze detection ratio & behavior
- Helps quickly identify **malicious or safe files**

Slip 11:

Q1. Using Metadefender for Malware Analysis

🛠 Steps to Scan File

1. Open Website

- Go to:
  👉 MetaDefender Cloud

---

2. Upload File

- Click **Upload File**
- Select suspicious binary (.exe, .dll)

---

3. Start Scan

- Wait for scanning to complete

---

4. View Results

- Check:
  - Detection results
  - Threat score
  - File details

🔍 What Metadefender Shows

- Multi-antivirus scan results
- File hash values
- File type & metadata
- Vulnerability detection
- Reputation score

⚖️ Comparison with VirusTotal

◈ Similarities

- Both use **multiple antivirus engines**
- Provide **hash-based search**
- Show detection results

◈ Differences

Feature	MetaDefender	VirusTotal
Focus	Security + vulnerability	Malware detection
Engines	Multiple AV engines	Larger AV community
Extra Features	Data sanitization, vulnerability scan	More behavioral reports
Interface	Simple & enterprise-focused	More detailed analysis

✅ Conclusion

- MetaDefender → good for **enterprise & vulnerability checks**
- VirusTotal → better for **detailed malware analysis**
- Use both for **better results**

Q2. Analyzing Malware in x64dbg

🛠 Steps to Open and Analyze

1. Open Malware File

- Launch x64dbg
- Click **File** → **Open** → **select .exe file**

---

2. Start Debugging

- Program pauses at **Entry Point**

---

3. Set Breakpoints

- Press **F2**
- Set breakpoints at:
  - Entry point
  - Suspicious functions

---

4. Run and Step Through

- Use:
  - **F7 (Step Into)**
  - **F8 (Step Over)**

🔍 Identify Key Indicators

⚠ 1. Anti-Debugging Techniques

Look for:

- API calls like:
  - IsDebuggerPresent()
  - CheckRemoteDebuggerPresent()

- Unusual jumps or crashes

👉 If found:

- Patch or bypass them

⚠ 2. Unpacking Routines

Signs:

- Packed file (small size, no readable strings)
- Code that:
  - Allocates memory (VirtualAlloc)
  - Writes data and jumps to new code

👉 Action:

- Dump memory after unpacking

⚠ 3. API Calls (Important)

Focus on:

- File APIs → CreateFile, WriteFile
- Network APIs → connect, send
- Process APIs → CreateProcess

⚠ 4. Suspicious Behavior

- File creation in temp folders
- Registry changes
- Network communication

□ Useful Features in x64dbg

- **CPU view** → instructions
- **Registers** → values
- **Memory dump** → data
- **Call stack** → execution flow

✅ Conclusion

- Use breakpoints + step execution
- Identify:
  - Anti-debugging
  - Unpacking
  - API calls
- Helps understand **malware behavior step-by-step**

Slip 12:

Q1. Using strings Command for Binary Analysis

🛠 What is strings?

- It extracts **readable text** from binary files
- Useful in **static malware analysis**

□ Example Command

strings sample.exe

👉 Save output to file:

strings sample.exe > output.txt

🔍 What Information You Can Extract

📌 1. URLs / Domains

- Example:
  - http://malicious-site.com

📌 2. IP Addresses

- Shows possible **command & control servers**

📌 3. File Paths

- Example:
  - C:\Windows\System32\...

📌 4. API Names

- Example:
  - CreateFile, WriteFile, connect

📌 5. Registry Keys

- Example:
  - Software\Microsoft\Windows\CurrentVersion\Run

📌 6. Error Messages / Debug Info

- Helps understand program behavior

📌 7. Suspicious Keywords

- Passwords, commands, hidden messages

⚠️ Why It Is Useful

- Quick overview of malware behavior
- No need to execute file
- Helps identify:
  - Network activity
  - File operations

✅ Conclusion

- strings is a **fast and simple tool**
- Helps extract important clues from malware

Q2. Loading and Analyzing DLL in IDA Pro

🛠️ Steps to Load DLL

1. Open IDA Pro

- Launch IDA

---

2. Open File

- Click **File** → **Open**
- Select .dll file

---

3. Auto Analysis

- IDA detects file type automatically
- Click **OK** and wait for analysis

🔍 Identify Entry Points

📌 1. DllMain Function

- Main entry point of DLL
- Handles:
  - Load
  - Unload events

📌 2. Exported Functions

- Open **Exports window**
- Shows functions used by other programs

📌 3. Entry Address

- Found in PE header
- Shows where execution begins

🔍 Analyze Key Functions

📌 1. Functions Window

- View all functions
- Look for suspicious names

📌 2. API Calls

Focus on:

- File → CreateFile, WriteFile
- Network → connect, send
- Registry → RegSetValue

📌 3. Strings Analysis

- Press **Shift + F12**
- Look for:
  - URLs
  - File paths
  - Commands

📌 4. Cross References (XREF)

- Press **X**
- Track function usage

📌 5. Imports Section

- Shows external libraries used
- Helps understand behavior

⚠️ Signs of Malicious DLL

- Hidden or obfuscated functions
- Network communication
- Auto-start registry entries
- Code injection behavior

✅ Conclusion

- Start with **DllMain + exports + strings**
- Analyze **functions, APIs, and flow**
- Helps understand **DLL behavior clearly**

Q1. Python Script for File Type & Hash Values

□ Python Script

```
import subprocess
import hashlib

file_path = input("Enter file path: ")

# Get file type using 'file' command
file_type = subprocess.getoutput(f"file {file_path}")
print("File Type:", file_type)

# Function to calculate hash
def calculate_hash(file_path, algo):
    hash_func = hashlib.new(algo)
    with open(file_path, "rb") as f:
        while chunk := f.read(4096):
            hash_func.update(chunk)
    return hash_func.hexdigest()

# Print hash values
print("MD5:", calculate_hash(file_path, "md5"))
print("SHA1:", calculate_hash(file_path, "sha1"))
print("SHA256:", calculate_hash(file_path, "sha256"))
```

🔍 Explanation (Simple)

- subprocess → runs Linux file command
- hashlib → generates hash values
- Reads file in chunks → efficient for large files
- Outputs:
  - File type
  - MD5, SHA1, SHA256 hashes

✅ Conclusion

- Script helps in **quick static analysis**
- Gives **file info + unique hash values**

Q2. Common Entry Points for Malware

⚠ **1. Email Attachments**

- Malware hidden in:
  - .exe, .pdf, .docx files

👉 Example:

- Fake job offer email with infected attachment

⚠ **2. Phishing Links (Social Engineering)**

- User clicks fake link
- Redirects to malicious website

👉 Example:

- Fake bank login page

⚠ **3. Software Vulnerabilities**

- Exploits bugs in software
- No user action needed

👉 Example:

- Outdated browser exploited by attacker

⚠ **4. Downloading from Untrusted Sites**

- Cracked software, pirated apps

👉 Example:

- Free software with hidden malware

⚠ 5. USB Devices (Removable Media)

- Infected USB spreads malware

👉 Example:

- Autorun malware

⚠ 6. Drive-by Downloads

- Visiting a website automatically downloads malware

⚠ 7. Network Attacks

- Malware spreads through network

👉 Example:

- Worm spreading in LAN

⚠ 8. Fake Software Updates

- User installs fake update

👉 Example:

- Fake Flash update

⚠ 9. Weak Passwords / RDP Attacks

- Attackers access system remotely

⚠ 10. Malicious Ads (Malvertising)

- Ads contain hidden malware

✅ Conclusion

- Malware enters through **user mistakes + system vulnerabilities**
- Common methods:
  - Email
  - Links
  - Downloads
- Awareness + security tools can **prevent attacks**

Slip 14:

Q1. Interpreting VirusTotal Results

🛠 Steps to Interpret Results

1. Check Detection Ratio

- Example:
  - **2/70** → likely safe (false positive possible)
  - **40/70** → likely malicious

👉 Higher detections = more dangerous

---

2. Check Malware Name

- Shows:
  - Trojan
  - Worm
  - Ransomware

👉 Helps identify malware type

---

3. Analyze File Details

- File type (.exe, .dll)
- Size
- Hash values

---

4. Behavior Analysis

- Check:
  - Network connections
  - File changes
  - Registry activity

---

5. Community Comments

- Security experts may provide insights

⚖️ Decision Making

✅ Likely Benign

- Very few detections (0–2)
- No suspicious behavior

⚠️ Suspicious

- Medium detections (3–10)
- Needs further analysis

🚑 Malicious

- High detections (10+)
- Clear harmful behavior

🔍 Next Steps if Malicious

◈ 1. Do Not Execute File

- Avoid running on main system

◈ 2. Perform Further Analysis

- Static analysis:
  - strings, IDA Pro
- Dynamic analysis:
  - VM + monitoring tools

◈ 3. Block Threat

- Block:
  - File hash
  - IPs/domains

◈ 4. Report Incident

- Inform security team
- Add to threat database

◈ 5. Remove File

- Delete or quarantine safely

⚠ Important Tip

- Even if 0 detections → file may still be **new malware**

✅ Conclusion

- Use detection ratio + behavior + comments
- Take action based on risk level
- Always verify with **multiple analysis methods**

Q2. Using Wireshark to Detect Phishing / Fake Websites

🛠 Steps to Analyze Network Traffic

1. Open Wireshark

- Start Wireshark
- Select network interface

---

2. Start Packet Capture

- Click **Start Capture**

---

3. Apply Filters

🔍 DNS Traffic

dns

🔍 HTTP Traffic

http

🔍 HTTPS (TLS)

tls

---

4. Analyze Packets

- Click packets and check:
    - Domain names

- IP addresses

🔍 Signs of Phishing / Fake Websites

⚠ 1. Suspicious Domain Names

- Look-alike domains
  👉 Example:
- paypa1.com instead of paypal.com

⚠ 2. Unknown IP Addresses

- Domains resolving to suspicious IPs

⚠ 3. No HTTPS / Invalid Certificate

- Unsafe connection

⚠ 4. Frequent Redirects

- Multiple redirects to different sites

⚠ 5. Strange DNS Requests

- Random or newly created domains

⚠ 6. Data Submission

- HTTP POST requests with sensitive data

▯ Useful Wireshark Features

- **Follow TCP Stream** → see full communication

- **Packet details pane** → inspect data
- **Statistics** → **Conversations** → track connections

✅ Conclusion

- Use filters + packet inspection
- Detect:
  - Fake domains
  - Suspicious traffic
- Helps identify **phishing attacks effectively**

Slip 15:

Q1. Configuring a VM for Dynamic Malware Analysis

🛠 Step-by-Step VM Configuration

1. Create Virtual Machine

- Use:
  - **VirtualBox / VMware**
- Install OS (Windows preferred for malware analysis)

---

2. Isolate the VM (Very Important)

- Set network to:
  - **Host-only** or **NAT (restricted)**
    👉 Prevent malware from spreading

---

3. Disable Sharing Features

- Turn OFF:

    - Shared folders
    - Clipboard sharing
    - Drag & drop

---

4. Take Snapshot

- Create snapshot before analysis
  👉 Restore system after malware execution

---

5. Install Guest Additions / VMware Tools

- For better performance (optional)

□ Key Tools to Install

🔍 **Monitoring Tools**

- Process Hacker
- Process Monitor (ProcMon)

🌐 **Network Analysis**

- Wireshark

□ Automation Tools

- Noriben (for logging malware activity)

□ Static Analysis Tools

- strings
- file
- hash tools (md5, sha256)

▯ Advanced Tools

- IDA Pro
- x64dbg

🛡 Other Useful Tools

- Fake DNS / INetSim (simulate internet)
- Regshot (track registry changes)

⚠ Extra Safety Tips

- Disable internet if not needed
- Use fake data inside VM
- Never use main system

✅ Conclusion

- VM must be **isolated + monitored**
- Install tools for:
  - Process
  - Network
  - File analysis
- Ensures **safe dynamic malware analysis**

Q2. Securing Network Against Phishing & Malicious Links

🛡 Layered Security Approach (Defense in Depth)

🛡 1. Email Security (First Layer)

- Use email filters to:
  - Block spam
  - Detect phishing emails
- Scan attachments automatically

🌐 2. Web Filtering

- Block access to:
  - Malicious websites
  - Fake domains

🔥 3. Firewall Protection

- Monitor and block:
  - Suspicious traffic
  - Unknown IPs

💻 4. Endpoint Protection

- Install antivirus/EDR on all systems
- Detect and block malware

🧠 5. User Awareness (Very Important)

- Train users to:
  - Avoid clicking unknown links
  - Verify emails
  - Check URLs

🔑 6. Multi-Factor Authentication (MFA)

- Adds extra security layer
- Prevents unauthorized access

📊 7. Network Monitoring

- Use tools like:
  - Wireshark
  - IDS/IPS

🔧 8. Patch Management

- Keep systems updated
- Fix vulnerabilities

🚫 9. Attachment Sandboxing

- Open suspicious files in sandbox/VM

🔍 10. URL Scanning

- Scan links before opening

⚠️ Examples of Protection

- Block phishing emails
- Detect fake login pages
- Stop malware download

✅ Conclusion

- Use **multiple layers of security**
- Combine:
  - Email filtering
  - Firewall
  - Endpoint protection
- Along with **user awareness** for best protection

Q1. Importance of VM Snapshots & Restoration

□ What is a Snapshot?

- A snapshot is a **saved state of a virtual machine**
- It stores:
  - OS state
  - files
  - settings

✅ Why Snapshots are Important

◈ 1. Safe Malware Testing

- Malware can damage system
- Snapshot allows safe testing

◈ 2. Quick Recovery

- Restore system in seconds
- No need to reinstall OS

◈ 3. Repeat Analysis

- Run malware multiple times

- Return to clean state each time

◈ 4. Prevent System Damage

- If malware crashes VM → restore easily

◈ 5. Save Time

- No need to create VM again

🛠 How to Create Snapshot

In VirtualBox / VMware:

1. Select VM
2. Click **Snapshots**
3. Click **Take Snapshot**
4. Give name (e.g., “Clean State”)

🔄 How to Restore Snapshot

1. Open VM
2. Go to **Snapshots**
3. Select saved snapshot
4. Click **Restore**
5. Confirm restore

👉 VM returns to original clean state

⚠ Important Tips

- Always take snapshot **before running malware**

- Do not store important data in VM

✅ Conclusion

- Snapshots ensure **safe, fast, and repeatable malware analysis**
- Easy recovery if anything goes wrong

Q2. Identifying Unusual Network Traffic in Wireshark

🛠 Steps to Analyze Traffic

1. Open Wireshark

- Select network interface

---

2. Start Capture

- Capture live traffic

---

3. Apply Filters

dns

http

tcp

🔍 Unusual Patterns to Look For

⚠ 1. High Traffic Volume

- Too many packets → possible malware activity

⚠ 2. Repeated Connections

- Same IP contacted repeatedly
  👉 May indicate **C2 (command & control)**

⚠ 3. Suspicious DNS Requests

- Random domain names
- Many failed requests (NXDOMAIN)

⚠ 4. Unknown IP Addresses

- Connections to unfamiliar or foreign IPs

⚠ 5. Data Exfiltration

- Large data being sent out of network

⚠ 6. Unusual Ports

- Traffic on uncommon ports

⚠ 7. Beaconing Behavior

- Regular interval communication
  ☞ Example: every 5 seconds

⚠ 8. Encrypted Suspicious Traffic

- Unknown TLS connections

▯ Useful Features

- **Follow TCP Stream** → see full communication
- **Statistics** → **Conversations** → track connections
- **Endpoints** → identify active devices

✅ Conclusion

- Look for:
  - Repetition
  - unknown connections
  - unusual traffic patterns
- Helps detect **infected systems in network**

Slip 17:

Q1. Precautions for Malware Analysis Lab Setup

⚠ 1. Isolation (Very Important)

- Keep lab **separate from main system/network**
- Use **Virtual Machines (VMs)**
  👉 Prevent malware from spreading

□ 2. Use Virtualization

- Tools:
  - VirtualBox
  - VMware
- Run malware only inside VM

🔄 3. Take Snapshots

- Take snapshot before testing
- Restore after analysis
  👉 Ensures clean system every time

🌐 4. Network Configuration

- Use:
  - **Host-only network** (most safe)
  - NAT (limited internet)
- Avoid direct internet access

🚫 5. Disable Sharing Features

- Turn OFF:
  - Shared folders
  - Clipboard
  - Drag & drop

🛡️ 6. Use Fake Environment

- Use dummy files, fake data
- Prevent real data theft

🔍 7. Install Monitoring Tools

- Process Hacker
- Process Monitor
- Wireshark

---

8. Use Analysis Tools

- IDA Pro
- x64dbg
- strings

🔐 9. Limit Permissions

- Do not run VM as admin unless needed

💾 10. Backup System

- Keep backup of clean VM

⚠️ 11. Disable Auto-Run

- Prevent automatic execution of malware

📝 12. Keep Logs

- Record all activities during analysis

🚫 13. Avoid Using Personal Files

- Never store important data in VM

✅ Conclusion

- Key precautions:
  - Isolation
  - Snapshots
  - Safe network
- Ensures **secure malware analysis environment**

Q2. Detecting Spam / Phishing Emails

🛠️ Steps to Detect Spam Email

1. Check Sender Email Address

- Look for:

- Fake or misspelled domains
  👉 Example: support@paypa1.com

---

2. Check Subject Line

- Urgent or threatening messages
  👉 Example:
- “Your account will be blocked!”

---

3. Analyze Email Content

- Poor grammar or spelling
- Unusual requests

---

4. Check Links Carefully

- Hover over link before clicking
- Verify real URL

---

5. Check Attachments

- Suspicious file types:
  - .exe, .zip, .docm

---

6. Use Email Client Features

- Spam filter
- “Mark as Spam” option

□ Using Third-Party Tools

🔍 Useful Tools

- VirusTotal → scan links/files
- Spamhaus → check domain reputation

🚨 Signs of Phishing Email

⚠ 1. Urgent Action Required

- “Click now or account will be blocked”

⚠ 2. Fake Login Pages

- Links to fake websites

⚠ 3. Request for Sensitive Data

- Asking:
  - Password
  - OTP
  - Bank details

⚠ 4. Mismatched URLs

- Link text ≠ actual link

⚠ 5. Unknown Attachments

- Malware hidden in files

⚠ 6. Generic Greetings

- “Dear Customer” instead of name

🛡️ Prevention Tips

- Do not click unknown links
- Verify sender
- Use antivirus & filters

✅ Conclusion

- Check:
  - Sender
  - links
  - content
- Use tools + awareness to detect **phishing emails**

Slip 18:

Q1. Installing & Using Process Hacker

🛠️ Steps to Install Process Hacker

1. Download Tool

- Go to official website:
  👉 Process Hacker

---

2. Download Installer

- Choose latest version (.exe file)

---

3. Install Application

- Run installer
- Click **Next** → **Install** → **Finish**

---

4. Run as Administrator

- Right-click → **Run as Admin**
  👉 Required for full access

⚙️ Basic Configuration

- Enable:
  - **Network tab**
  - **Process tree view**
- Adjust columns:
  - CPU, Memory, PID

🔍 Essential Features for Malware Analysis

🔹 1. Process Tree

- Shows parent-child processes
  👉 Helps detect suspicious spawning

🔹 2. Real-Time Monitoring

- View CPU, memory usage
  👉 Identify abnormal activity

🔹 3. Process Properties

- Right-click → Properties
- Check:
  - File path
  - Threads

    - Handles

◈ 4. Network Activity

- Shows active connections
  ☞ Detect malware communication

◈ 5. DLLs & Handles

- See loaded DLL files
  ☞ Identify injected code

◈ 6. Kill / Suspend Process

- Stop suspicious processes safely

◈ 7. Search Online

- Check process reputation

⚠ Why It Is Useful

- Detect:
    - Hidden processes
    - Malware behavior
    - System changes

☑ Conclusion

- Process Hacker is a **powerful monitoring tool**
- Helps in **real-time malware analysis**

Q2. How Email Filtering Works

📧 What is Email Filtering?

- Process of **detecting and blocking spam/phishing emails**

🛠️ How Email Filtering Works

1. Incoming Email Check

- Email server scans every incoming mail

---

2. Analyze Content & Sender

- Check:
  - Email address
  - Message content
  - Attachments

---

3. Apply Filtering Rules

- Based on predefined rules

---

4. Categorize Email

- Spam / Legitimate / Suspicious

🔍 Common Techniques Used

🔹 1. Blacklist Filtering

- Blocks known spam email addresses/domains

🔹 2. Whitelist Filtering

- Allows trusted senders

◈ 3. Keyword Filtering

- Detects spam words:
  - “Free”, “Win money”, “Urgent”

◈ 4. Heuristic Analysis

- Detects suspicious patterns
- Example:
  - Too many links
  - Fake content

◈ 5. Machine Learning

- Learns from user behavior
- Improves spam detection

◈ 6. Attachment Scanning

- Checks files for malware

◈ 7. URL Analysis

- Scans links for phishing

◈ 8. Sender Authentication

- Uses:
  - SPF
  - DKIM

    - DMARC

⚠️ Benefits

- Blocks spam emails
- Prevents phishing attacks
- Protects user data

✅ Conclusion

- Email filtering uses **multiple techniques**
- Combines:
    - Rules
    - AI
    - Security checks
- Ensures safe email communication

Slip 19:

Q1. Using Process Hacker to Analyze Malware

🛠️ Steps to Analyze Malware

1. Run Process Hacker

- Open tool as **Administrator**

---

2. Start Monitoring

- Keep Process Hacker open
- Run malware sample inside VM

---

3. Observe New Processes

- Look for:
  - Unknown processes
  - Suspicious names

---

4. Check Process Tree

- Identify:
  - Parent → Child processes
    👉 Malware often creates multiple child processes

---

5. Analyze Process Properties

- Right-click process → Properties
- Check:
  - File location
  - Command line

🔍 Key Indicators to Focus On

🔹 1. Processes

- Unknown or hidden processes
- High CPU or memory usage

🔹 2. Threads

- Multiple threads running
  👉 Indicates complex/malicious activity

🔹 3. Modules (DLLs)

- Check loaded DLL files

- Suspicious or unknown DLLs

◈ 4. Handles

- Access to:
  - Files
  - Registry
    ☞ Malware interacts with system resources

◈ 5. Network Activity

- Outgoing connections
- Unknown IP addresses

◈ 6. Process Behavior

- Frequent creation/termination
- Process injection

◈ 7. Digital Signature

- Unsigned or fake signatures

⚠ Suspicious Signs

- Random process names
- Running from unusual folders (Temp, AppData)
- Multiple child processes

☑ Conclusion

- Focus on:

    - Processes
    - Threads
    - Modules
- Helps detect **real-time malware behavior**

Q2. Phishing Attack Techniques & Indicators

🎯 **Common Phishing Techniques**

🔹 **1. Email Phishing**

- Fake emails pretending to be trusted organizations

🔹 **2. Spear Phishing**

- Targeted attack on specific person/company

🔹 **3. Link Manipulation**

- Fake URLs that look real

🔹 **4. Fake Websites**

- Clone of real website login page

🔹 **5. Attachment-Based Attacks**

- Malicious files in email

🔹 **6. Social Engineering**

- Trick user using urgency or fear

🕵 Key Indicators of Phishing

⚠ 1. Suspicious Sender Email

- Misspelled or unknown domain

⚠ 2. Urgent or Threatening Message

- “Account will be blocked”

⚠ 3. Fake Links

- URL mismatch
  👉 Hover to check real link

⚠ 4. Request for Sensitive Data

- Asking for:
  - Password
  - OTP
  - Bank details

⚠ 5. Poor Grammar & Spelling

- Common in fake emails

⚠ 6. Unknown Attachments

- .exe, .zip, .docm files

⚠ 7. No HTTPS / Invalid Certificate

- Unsafe websites

⚠️ 8. Generic Greeting

- “Dear User” instead of name

🛡️ Prevention Tips

- Do not click unknown links
- Verify sender
- Use antivirus and email filters

✅ Conclusion

- Phishing uses **tricks + fake identity**
- Check:
  - sender
  - links
  - content
- Helps avoid **cyber attacks**

Slip 20:

Q1. Monitoring Malware Using Process Hacker

🛠️ Steps to Monitor System Activity

1. Open Process Hacker

- Run as **Administrator**

---

2. Start in Clean VM

- Take snapshot

- Ensure system is clean

---

3. Run Malware Sample

- Execute inside VM
- Observe changes in Process Hacker

🔍 What to Monitor

◈ 1. Process Creation

- Look for:
  - New processes
  - Unknown names
- Use **Process Tree**
  👉 Malware creates child processes

◈ 2. File System Activity

- Check:
  - File paths in process properties
- Look for:
  - Files created in Temp/AppData

◈ 3. Registry & Handles

- Check **Handles tab**
  👉 Malware may modify registry or system files

◈ 4. Network Connections

- Open **Network tab**

- Look for:
  - Unknown IPs
  - Suspicious connections

◈ 5. CPU & Memory Usage

- High usage = suspicious activity

◈ 6. DLL / Modules

- Check loaded DLLs
- Identify unknown or injected modules

◈ 7. Threads

- Many threads → complex behavior

⚠ Suspicious Indicators

- Random process names
- Processes running from unusual locations
- Frequent process creation
- External network connections

▯ Extra Tips

- Compare before vs after running malware
- Use snapshot to restore system

✅ Conclusion

- Monitor:
  - Processes
  - files
  - network
- Helps understand **malware behavior in real-time**

Q2. Protecting from Phishing Using Browser Tools

🛡️ **Using Browser Extensions / Security Tools**

🔧 **Examples:**

- Google Safe Browsing
- Netcraft Extension

🛠️ **How These Tools Work**

1. Website Reputation Check

- Compares website with known phishing database

---

2. URL Analysis

- Detects:
  - Fake or suspicious URLs

---

3. Warning Alerts

- Shows warning before opening dangerous site

---

4. Real-Time Protection

- Blocks harmful websites automatically

---

5. Certificate Checking

- Verifies HTTPS security

🕵️ How Tools Identify Fraudulent Sites

⚠️ 1. Fake Domain Names

- Example:
  - amaz0n.com instead of amazon.com

⚠️ 2. Blacklisted Websites

- Known phishing sites are blocked

⚠️ 3. Suspicious Page Behavior

- Fake login forms
- Data stealing attempts

⚠️ 4. SSL Certificate Issues

- Missing or invalid certificate

⚠️ 5. Page Content Analysis

- Detects phishing patterns

🛡️ Extra Safety Tips

- Always check URL manually
- Avoid clicking unknown links
- Keep browser updated

✅ Conclusion

- Browser tools provide:
  - Real-time alerts
  - Website verification
- Help prevent **phishing attacks effectively**

Slip 21:

Q1. Installing & Configuring Process Monitor

🛠 Steps to Install Process Monitor

1. Download Tool

- Go to official Microsoft Sysinternals site
  👉 Process Monitor

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2. Extract ZIP File

- Download comes as .zip
- Extract using WinRAR/Zip

---

3. Run Application

- Open Procmon.exe
- Run as **Administrator**

⚙ Important Configuration Settings

◈ 1. Stop Default Capture

- Click **Capture (Ctrl + E)** to pause
  ☞ Prevent unnecessary logs

◈ 2. Clear Old Logs

- Click **Clear (Ctrl + X)**
  ☞ Start fresh

◈ 3. Set Filters (Very Important)

Example Filters:

- Process Name = malware.exe
- Operation = CreateFile / RegSetValue

☞ Reduces noise and focuses on malware

◈ 4. Enable Required Events

- File System Activity ✔
- Registry Activity ✔
- Process/Thread Activity ✔
- Network Activity ✔

◈ 5. Use Highlighting

- Highlight suspicious operations

◈ 6. Enable Boot Logging (Optional)

- Capture early system activity

🔍 During Malware Analysis

Monitor:

- File creation/modification
- Registry changes
- Process creation
- DLL loading

⚠️ Tips

- Use VM + snapshot
- Avoid too many logs (use filters)

✅ Conclusion

- Configure filters + capture settings
- Helps track **malware behavior in detail**

Q2. Responding to Phishing Email

📥 If You Receive a Suspicious Email

🔹 1. Do Not Click Anything

- Avoid:
  - Links
  - Attachments

🔹 2. Verify Sender

- Check email address carefully
- Contact bank/employer using official website

◈ 3. Check for Phishing Signs

- Urgent message
- Fake links
- Grammar mistakes

◈ 4. Report the Email

- Use email option:
  - “Report Phishing”
- Inform IT/security team

◈ 5. Delete the Email

- Remove from inbox

◈ 6. Scan with Security Tools

- Use:
  - VirusTotal (for links/files)

⚠ If You Clicked by Mistake

◆ 1. Change Password Immediately

- Bank/email accounts

◆ 2. Enable MFA

- Add extra security

◆ 3. Run Antivirus Scan

- Check system for malware

◆ 4. Inform Bank/Company

- Report possible compromise

🛡 Prevention Tips

- Always check URL
- Do not trust unknown emails
- Keep system updated

✅ Conclusion

- Never trust suspicious emails
- Verify, report, and delete
- Quick action prevents **data theft and fraud**

Slip 22:

Q1. Filtering Noise in Process Monitor

🛠 Using Process Monitor Effectively

🔇 Why Filtering is Needed

- Process Monitor captures **too many events**
- Filtering helps focus only on **malware activity**

⚙ Steps to Filter Noise

◈ 1. Stop Capture First

- Press **Ctrl + E**
  ☞ Pause logging

◈ 2. Clear Old Logs

- Press **Ctrl + X**
  ☞ Start fresh

◈ 3. Apply Filters

Go to **Filter** → **Filter (Ctrl + L)**

Common Filters:

- Process Name **is** malware.exe → Include
- Process Name **is not** system processes → Exclude
- Operation **is** CreateFile / WriteFile → Include

◈ 4. Exclude System Noise

- Remove:
  - svchost.exe
  - explorer.exe
  - other normal processes

◈ 5. Use Highlighting

- Highlight suspicious events
  ☞ Easy to identify

◈ 6. Start Capture Again

- Press **Ctrl + E**
- Run malware sample

🔍 Important Events to Look For

◈ 1. File System Activity

- CreateFile
- WriteFile
  ☞ File creation/modification

◈ 2. Registry Activity

- RegCreateKey
- RegSetValue
  ☞ Persistence (startup entries)

◈ 3. Process Activity

- Process Create
- Thread Create
  ☞ New processes

◈ 4. DLL Loading

- Load Image
  ☞ Suspicious DLLs

◈ 5. Network Activity

- TCP/UDP connections

⚠ Suspicious Indicators

- Activity in:
  - Temp folder
  - AppData
- Repeated registry changes
- Unknown processes

🔹 Tips

- Use multiple filters together
- Save logs for analysis

✅ Conclusion

- Filtering removes noise
- Focus on:
  - files
  - registry
  - processes
- Helps in **accurate malware analysis**

Q2. Blocking Unauthorized USB Devices

🛡 Goal

- Allow only **approved USB devices**
- Block unknown devices

⚙ Methods to Configure USB Security

◈ 1. Use Group Policy (Windows)

Steps:

1. Press **Win + R** → **gpedit.msc**
2. Go to:
    - Computer Configuration
    - Administrative Templates
    - System → Removable Storage Access
3. Enable:
    - “Deny all removable storage access”

◈ 2. Allow Approved Devices

- Use **Device Installation Restrictions**
- Allow only specific device IDs

◈ 3. Use Device Manager

- Disable USB ports manually
- Or disable unknown devices

◈ 4. Use Security Software

- Endpoint protection tools
- Control USB usage

◈ 5. Registry Method (Advanced)

- Disable USB via registry settings

🔹 Device Management Policy

Include:

- Allow only trusted USB devices
- Block unknown devices
- Log USB usage
- Require admin approval

🔍 Additional Controls

- Enable alerts for USB usage
- Scan USB devices automatically

⚠️ Benefits

- Prevents:
  - Malware via USB
  - Data theft

🛡️ Example Policy

- Company USB only
- Personal USB blocked
- Activity logged

✅ Conclusion

- Use:
  - Group Policy
  - Device control
- Ensures **secure USB access management**

Slip 23:

Q1. Using Process Monitor to Detect File & Registry Changes

🛠 Tool Used

- Process Monitor

⚙ Steps to Identify Changes

◈ 1. Start Process Monitor

- Run as **Administrator**

◈ 2. Stop & Clear Logs

- Press **Ctrl + E** (stop)
- Press **Ctrl + X** (clear)

◈ 3. Apply Filters

- Filter by:
  - Process Name = malware.exe
    👉 Focus only on malware activity

◈ 4. Start Capture

- Press **Ctrl + E**
- Run malware sample

◈ 5. Monitor Events

Look for:

🗂 File System Changes

- **CreateFile** → new file created
- **WriteFile** → file modified

🗃 Registry Changes

- **RegCreateKey** → new registry key
- **RegSetValue** → value modified

⚙ Process Activity

- Process Create
- Load Image (DLL loading)

🔍 Example Event to Investigate

👉 RegSetValue Event

Why Important:

- Malware may add itself to startup

Example Path:

- HKCU\Software\Microsoft\Windows\CurrentVersion\Run

👉 This means:

- Malware runs automatically on system startup

⚠ Suspicious Indicators

- Changes in:
  - System folders
  - Startup registry keys

- Files created in:
  - Temp
  - AppData

▯ Tips

- Use filters to reduce noise
- Save logs for analysis

✅ Conclusion

- Focus on:
  - File changes
  - Registry modifications
- Helps detect **malware persistence and behavior**

Q2. Safely Analyzing a Suspicious USB Drive

⚠ Goal

- Analyze USB **without infecting main system**

🛠 Safe Analysis Steps

◈ 1. Use Isolated System

- Use:
  - Virtual Machine (VM)
    👉 No connection to main network

◈ 2. Disable Auto-Run

- Prevent automatic execution of malware

◈ 3. Do Not Open Files Directly

- Avoid double-clicking unknown files

◈ 4. Scan USB First

Use Tools:

- Windows Defender
- VirusTotal

◈ 5. Check File Types

- Look for:
  - .exe, .bat, .vbs
    ☞ Suspicious files

◈ 6. Use Static Analysis Tools

- strings → extract text
- hash tools → check file identity

◈ 7. Monitor with Tools

- Process Monitor
- Process Hacker

◈ 8. Analyze in Sandbox

- Run suspicious file inside VM
- Observe behavior

🔍 Techniques to Detect Malware

⚠ Hidden Files

- Enable “Show hidden files”

⚠ Fake Shortcuts

- .lnk files pointing to malware

⚠ Autorun Files

- Check autorun.inf

⚠ Unknown Executables

- Random names or unusual size

🛡 Extra Safety Tips

- Disconnect internet
- Take VM snapshot
- Do not copy files to main system

✅ Conclusion

- Always use **isolated environment**
- Scan + analyze before opening
- Prevents **infection of main network**

Slip 24:

Q1. Noriben Tool (Dynamic Malware Analysis)

▯ What is Noriben?

- Noriben
- It is a **Python-based script** that uses:
  - Process Monitor

👉 It automates malware behavior logging

🔄 Difference from Other Tools

Feature	Noriben	Process Monitor	Process Hacker
Type	Automation tool	Monitoring tool	Monitoring tool
Usage	Automatic logging	Manual analysis	Real-time monitoring
Output	Clean report	Raw logs	Live view
Ease	Easy	Complex	Moderate

👉 Noriben = **automatic + clean report**
👉 Others = **manual analysis**

🛠 Steps to Install Noriben

◈ 1. Install Python

- Required to run Noriben

◈ 2. Download Noriben

- Get script from official GitHub

◈ 3. Install Process Monitor

- Required dependency

◈ 4. Extract Files

- Place Noriben files in folder

◈ 5. Run Noriben

python Noriben.py

⚙ How Noriben Works

1. Starts Process Monitor
2. Runs malware sample
3. Captures activity
4. Generates report

📊 Output Includes

- File changes
- Registry changes
- Network activity

⚠ Advantages

- Easy to use
- Automated logging
- Clean readable report

✅ Conclusion

- Noriben simplifies **dynamic malware analysis**

- Best for beginners

Q2. Setting up REMnux VM

□ What is REMnux?

- REMnux
  👉 Linux-based toolkit for malware analysis

🛠 Installation Steps

◈ 1. Download REMnux

- Download OVA file (pre-built VM)

◈ 2. Install Virtualization Tool

- VirtualBox / VMware

◈ 3. Import VM

- Open VirtualBox/VMware
- Select **Import Appliance**
- Choose REMnux .ova file

◈ 4. Configure Resources

- RAM: 4GB+
- CPU: 2 cores

◈ 5. Start VM

- Boot the REMnux system

⚙ Post-Installation Setup

◈ 1. Update System

remnux upgrade

◈ 2. Configure Network

- Use:
  - Host-only or NAT

◈ 3. Verify Tools

- Check tools like:
  - Wireshark
  - Volatility
  - INetSim

◈ 4. Take Snapshot

- Save clean state

◈ 5. Disable Internet (if needed)

- For safe malware testing

□ Why REMnux is Useful

- Pre-installed tools
- Ready for malware analysis
- Saves setup time

⚠ Safety Tips

- Always use VM
- Do not connect to main network

✅ Conclusion

- REMnux is a **complete malware analysis environment**
- Easy setup using OVA + ready tools

Slip 25:

Q1. Role of rundll32.exe in Windows

□ What is rundll32.exe?

- rundll32.exe
- A **Windows system file** used to run functions from DLL files

👉 DLL files cannot run directly, so rundll32.exe helps execute them

---

🛠 How It Executes a DLL

Basic Syntax:

rundll32.exe filename.dll,FunctionName

Example:

rundll32.exe example.dll,StartFunction

👉 It loads the DLL and runs the specified function

⚠ Why Malware Uses rundll32.exe

◈ 1. Hides Malicious Activity

- Looks like a **legitimate Windows process**

◈ 2. Bypass Security Detection

- Antivirus may trust system processes

◈ 3. Execute Malicious DLLs

- Run hidden payload inside DLL

◈ 4. Fileless Attacks

- No need for .exe file

◈ 5. Persistence

- Can be used in:
  - Startup registry
  - Scheduled tasks

🕵 Suspicious Indicators

- rundll32.exe running from unusual location
- Unknown DLL file
- Strange command line arguments

🛡 Example of Malicious Use

- Malware runs:
  - rundll32.exe + hidden DLL
    👉 Executes payload silently

✅ Conclusion

- rundll32.exe is **legitimate but can be misused**
- Common technique for **stealthy malware execution**

Q2. Static Analysis Using REMnux

🧰 Tool Used

- REMnux

👉 Linux toolkit for malware analysis

🛠 Steps for Basic Static Analysis

🔹 1. Transfer Malware Sample

- Copy file into REMnux VM
- Do NOT execute

🔹 2. Identify File Type

Command:

```
file sample.exe
```

👉 Shows file type (PE, ELF, etc.)

🔹 3. Generate Hash Values

Commands:

```
md5sum sample.exe
sha1sum sample.exe
sha256sum sample.exe
```

👉 Used for identification

◈ 4. Extract Strings

Command:

strings sample.exe

👉 Finds:

- URLs
- IP addresses
- File names

◈ 5. Check File Details

- Use tools like:
  - readelf
  - objdump

◈ 6. Disassemble File

- Use tools:
  - radare2
  - objdump

👉 View low-level code

🔧 Tools in REMnux

🔍 File Identification

- file

🔑 Hashing

- md5sum
- sha1sum
- sha256sum

📄 String Analysis

- strings

⬜ Disassembly

- radare2
- objdump

⚠️ Safety Tips

- Never execute file
- Work inside VM only

✅ Conclusion

- REMnux provides tools for:
  - File analysis
  - Hashing
  - Disassembly
- Helps understand malware **without running it**